EUNOIA JUNIOR COLLEGE
JC2 Preliminary Examination 2021
General Certificate of Education Advanced Level
Higher 2

H2 Biology

9744/01

Paper 1 Multiple Choice

22 September 2021

60 minutes

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use paper clips, glue or correction fluid.
Write your name, civics group and registration number on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

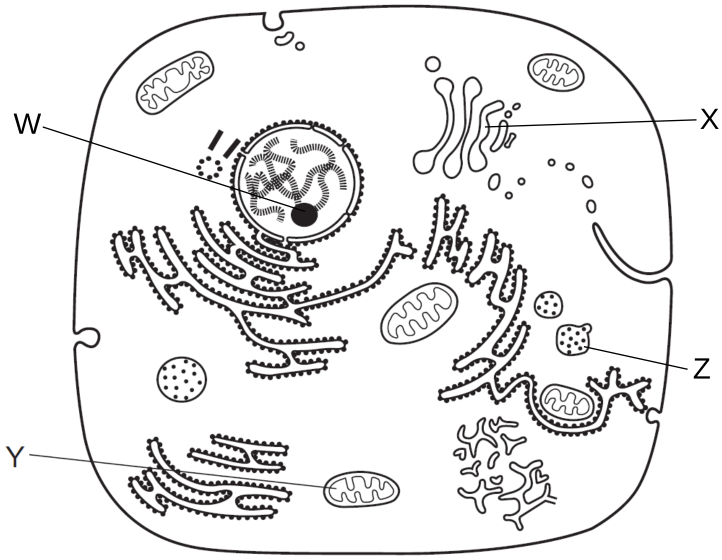
Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **21** printed pages and **1** blank page.

Answer **all** questions.

1 The diagram below shows a eukaryotic cell with four organelles labelled W to Z.



Which of the following rows correctly matches the organelle to its function?

	W	X	Y	Z
A	site of assembly of rRNA with proteins	glycosylation of cell surface receptors	oxidation of fatty acids and glucose	transport of proteins to be secreted
B	site of active gene transcription	synthesis and storage of lipids	site of Krebs cycle reactions	synthesis of cytosolic proteins
C	site of attachment of kinetochore proteins	sorting of proteins into vesicles	site of Calvin cycle reactions	transport of proteins to be modified
D	site of synthesis of ribosomal proteins	packaging of lysosomal enzymes	storage of glucose as glycogen	breakdown of foreign particles

2 A student carried out the Benedict's test on three different solutions, X, Y and Z. All three solutions have the same concentration, but contain different carbohydrates.

A sample of each solution was incubated with acid for five minutes, and then heated in a boiling water-bath with Benedict's solution. The time taken for the first appearance of a colour change was recorded.

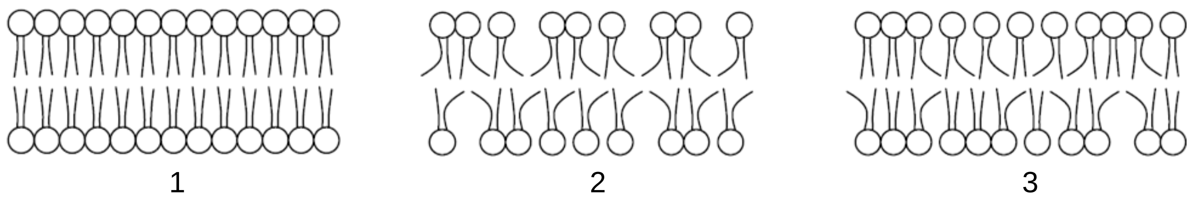
The results are shown in the table below.

solution	time for first appearance of a colour change with Benedict's solution / s
X	14
Y	300
Z	28

Which row correctly identifies the carbohydrate present in each solution?

	X	Y	Z
A	glucose	starch	maltose
B	glucose	sucrose	lactose
C	lactose	starch	glucose
D	lactose	sucrose	maltose

3 Cell membranes in three different organisms adapted for different climates are shown below.



Several regions of the world are known to have the following climates:

- desert region is hot and dry
- tropical region is warm and wet
- temperature region is cool and moderate
- arctic region is cold and dry

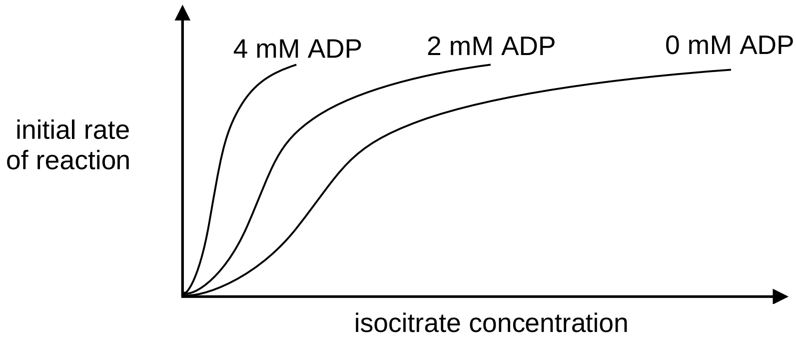
Which of the following correctly matches the regions to each type of cell membrane?

	1	2	3
A	desert	arctic	temperate
B	arctic	desert	tropical
C	desert	temperate	arctic
D	tropical	arctic	desert

4 Isocitrate dehydrogenase catalyses the following reaction in the Krebs cycle:

$$\text{isocitrate} + NAD^+ \rightarrow \alpha\text{-ketoglutarate} + CO_2 + NADH$$

The curves in the graph below show the initial rate of reaction against increasing isocitrate concentration in the presence of various levels of ADP and excess NAD^+.



Which statement correctly describes the effect of ADP on isocitrate dehydrogenase?

A ADP competes with isocitrate for the active site of isocitrate dehydrogenase.

B ADP binds to a site other than the active site of isocitrate dehydrogenase, and induces a change of its 3D conformation.

C ADP binds to an allosteric site of isocitrate dehydrogenase and prevents binding of isocitrate to the active site.

D ADP binds to an allosteric site of isocitrate dehydrogenase and makes it easier for isocitrate to bind to the active site.

5 A number of blood transfusion laboratories around the world are hoping to use embryonic stem cells to produce large numbers of red blood cells.

Which statement correctly describes the differentiation of embryonic stem cells to red blood cells?

A Appropriate molecular signals are required to induce embryonic stem cells to differentiate into pluripotent blood stem cells and then into red blood cells.

B Appropriate molecular signals are required to induce totipotent embryonic stem cells to differentiate into blood stem cells and then into red blood cells.

C Appropriate chemical signals are required to induce embryonic stem cells to undergo differential gene expression to form red blood cells.

D Appropriate chemical signals are required to induce multipotent embryonic stem cells to undergo differential gene expression to form red blood cells.

6 A space probe brought back some living material from a distant planet. Proteins very similar to those found on Earth were present.

DNA of this living material, which exists as double helices, contains six different nucleic acid bases, identified as H, I, J, K, L and M. Among these, J, L and M were found to be pyrimidine bases.

Studies of the DNA from this living material gave the following results:

ratio of bases in double-stranded DNA	numerical value
H / J	1.00
(H + I) / (J + M)	1.02
(H + K) / (J + M)	1.14
(H + K) / (J + L)	1.01

Assuming that complementary base pairing only occurs between a purine and a pyrimidine, which of the following are valid conclusions from these data?

1 H base pairs with J.

2 H, I, K are likely to be purine bases.

3 It is probable that I base pairs with M, and K base pairs with L.

A 1 and 2

B 1 and 3

C 2 and 3

D 1, 2 and 3

7 The table below shows the anticodon sequences for some amino acids.

amino acid	anticodon (3' to 5')
alanine (A)	CGG / CGU
cysteine (C)	ACG
phenylalanine (F)	AAG
threonine (T)	UGU / UGA

Part of a polypeptide chain has the sequence shown below.

$$^{-}OOC-T-C-A-F-A-NH_3^{+}$$

Which of the following shows the correct sequence of nucleotides in the template DNA strand that codes for this section of the polypeptide chain?

A 5' – AGT GCA GGC GAA TGC – 3'

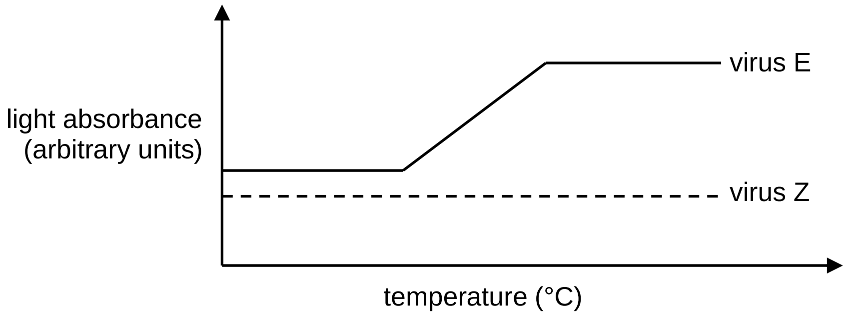
B 5' – CGT AAG CGG ACG TGA – 3'

C 5' – TGA ACG CGT AAG CGG – 3'

D 5' – TGC GAA GGC GCA AGT – 3'

8 Two new viruses E and Z, which infect eukaryotic cells, have been identified.

In one experiment, the nucleic acid from each virus is isolated and analysed over a range of temperatures. The light absorbance of nucleic acids increases when denaturation occurs. The behaviour of the nucleic acid from each virus is shown in the graph.



In a second experiment, it is found that treatment with reverse transcriptase inhibitors or with inhibitors of DNA synthesis blocks the ability of virus Z to infect cells. In contrast, reverse transcriptase inhibitors have no effect on the ability of virus E to infect cells but DNA synthesis inhibitors block infection by virus E.

Which of the following conclusions can be drawn from the results of both experiments?

A The genome of virus E is single-stranded RNA and that of virus Z is double-stranded DNA.

B The genome of virus E is double-stranded DNA and that of virus Z is single-stranded RNA.

C The genome of virus E is double-stranded RNA and that of virus Z is single-stranded DNA.

D The genome of virus E is double-stranded DNA and that of virus Z is double-stranded RNA.

9 The end result of conjugation between two bacterial cells

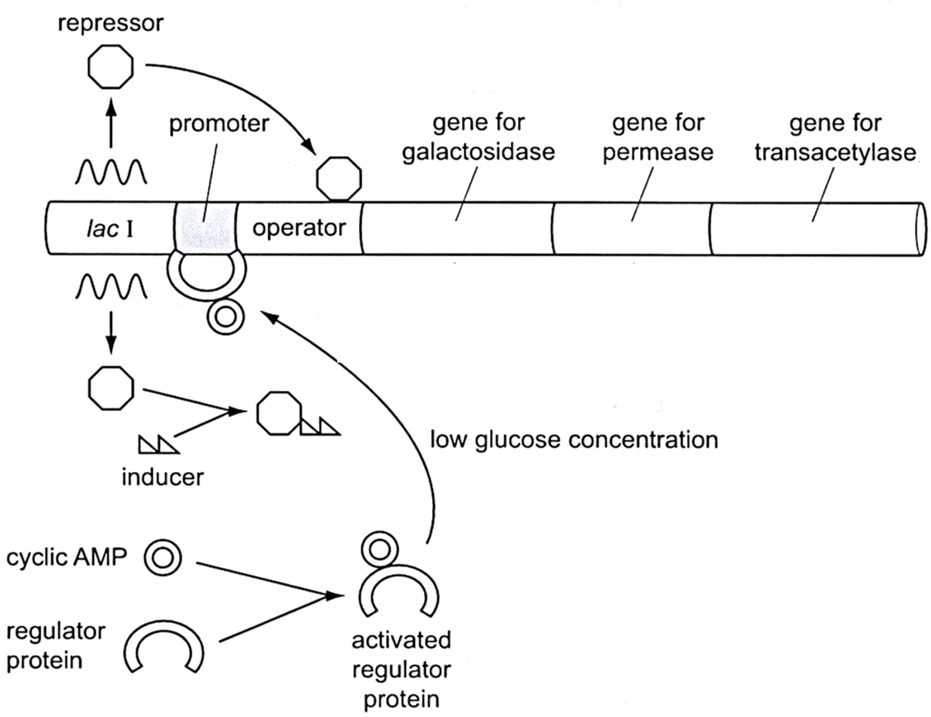
A results in two F^+ strains.

B involves transfer of the entire bacterial chromosome.

C converts the recipient strain to F^+ and the donor to F^-.

D permanently link them via a mating bridge.

10 The diagram shows some of the factors affecting the *lac* operon.



Which statement describes the role of cyclic AMP in the expression of the *lac* operon?

A It activates the inducer to bind to the *lac* operator.

B It activates a regulator so RNA polymerase binds more effectively to the promoter.

C It prevents the expression of the *lac* operon when the glucose concentration is high.

D It prevents the regulator protein binding to the promoter when the glucose concentration is high.

11 Which of the following is **not** a feature of eukaryotic gene expression?

A Polycistronic mRNAs are very rare.

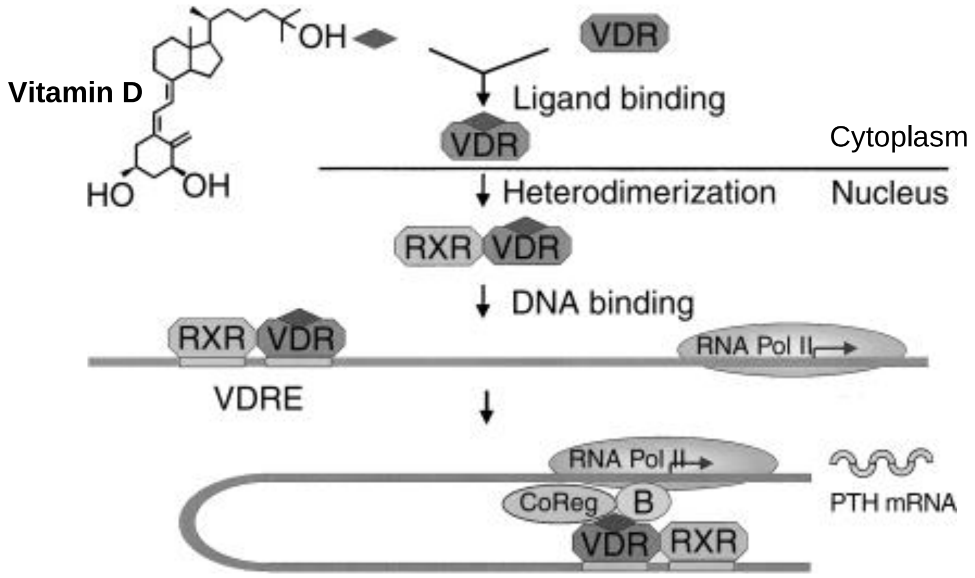
B Many genes are interrupted by non-coding DNA sequences.

C RNA synthesis and protein synthesis are coupled.

D mRNA is often extensively modified before translation.

12 Vitamin D helps the body to increase its calcium concentration for the production of strong bones.

The figure below illustrates the effect of vitamin D binding to the vitamin D receptor (VDR).



The parathyroid hormone (PTH) increases calcium concentration in the body by increasing calcium absorption in the intestine and by increasing calcium reabsorption in the kidney.

Which of the following statement(s) is/are true?

1 VDR is an intracellular receptor for Vitamin D.
2 RXR-VDR-Vitamin D complex functions as an enhancer to bring about an increase in the expression of PTH.
3 Vitamin D brings about an increase in the transcription of *PTH* gene.

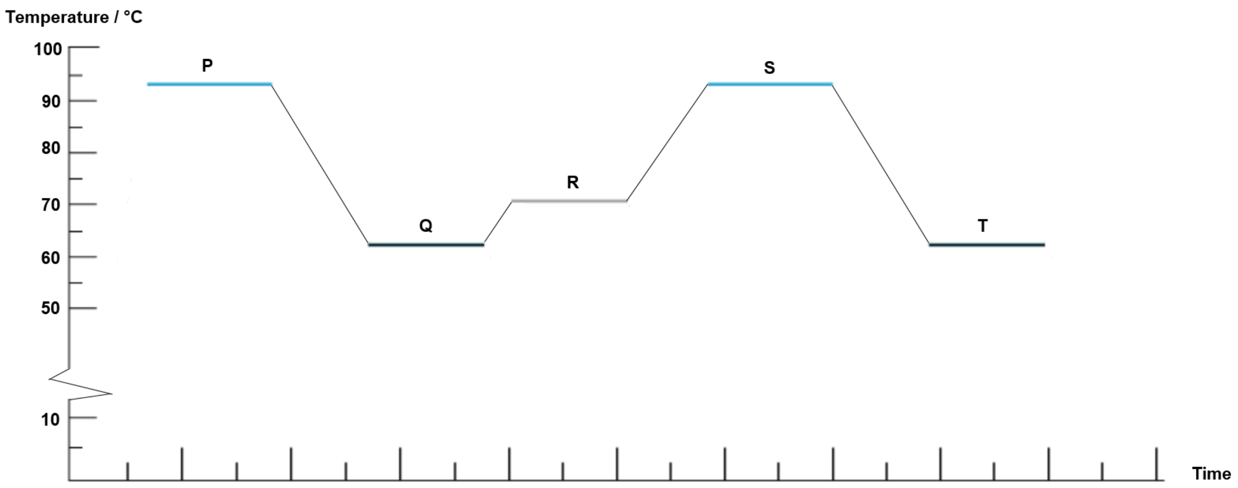
A 1 only

B 1 and 3 only

C 2 and 3 only

D 1, 2 and 3

13 PCR is carried out with repeated cycles of temperature changes in a thermocycler machine. The diagram below shows part of the cycles of temperature used during the amplification of a target gene within a bacterial DNA sample.



Based on the information above, how many of the following statement(s) is/are **incorrect**?

- Carrying out a cycle of temperatures P, Q and R twice will result in an increase in the amount of DNA by 4 times.
- Hydrogen bonds are formed when temperatures R and T are applied.
- RNA primers are usually provided in excess during PCR to increase the chance of the primers annealing to the template DNA.
- High temperatures such as P and S are required as *Taq* DNA polymerase is derived from a bacteria that was found in hot springs.

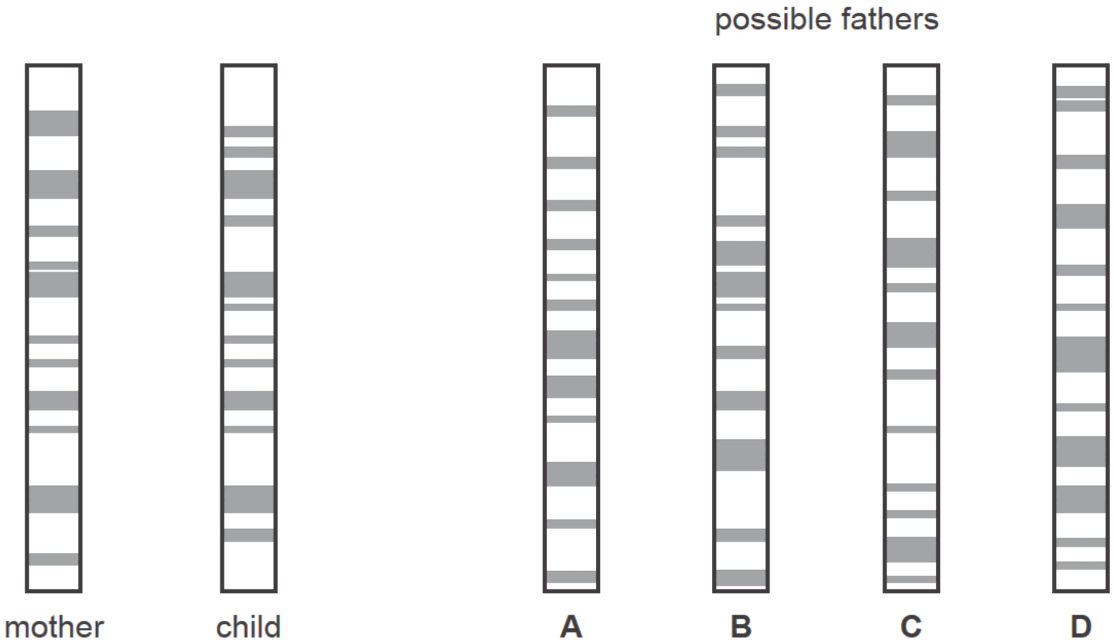
A 1 **B** 2 **C** 3 **D** 4

14 Genetic profiling can be used to determine the paternity of a child.

DNA from the mother and the child is cut into fragments, separated by electrophoresis and made visual using a stain.

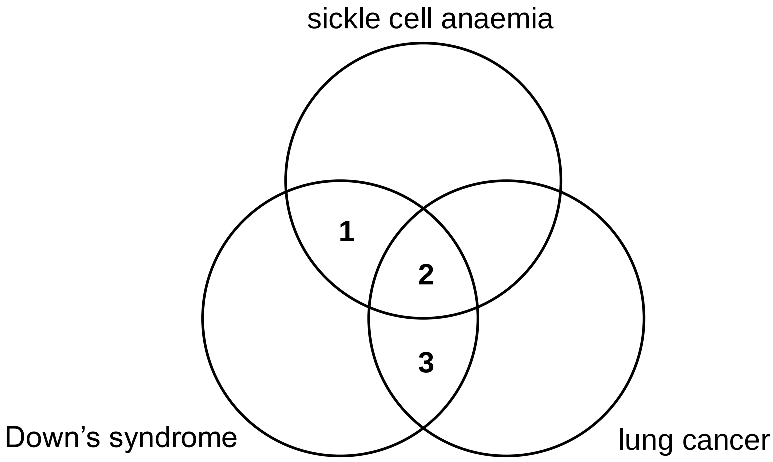
The diagram shows the genetic profiles of a mother and child, and four possible fathers.

Who is the father?



Ans = **B**

15 Features of three diseases are identified with a number in the diagram below.



Which combination correctly describes these features?

	1	**2**	**3**
A	has a hereditary genetic component	may be caused by chromosomal translocation	can arise due to environmental causes
B	only arise due to genetic abnormalities	has a hereditary genetic component	may be caused by chromosomal translocation
C	can arise due to environmental causes	only arise due to genetic abnormalities	may be caused by chromosomal translocation
D	only arise due to genetic abnormalities	has a hereditary genetic component	can arise due to environmental causes

16 During telophase of mitosis, a scientist stains the chromosomes of a diploid animal cell with a fluorescent dye to observe the telomeres. This cell has a diploid number of 22.

How many telomeres will the scientist observe in one of the nuclei?

A 11 **B** 22 **C** 44 **D** 88

17 Huntington's disease is a genetic disorder that results in the degeneration of the nervous system.

Several discoveries were made regarding Huntington's disease.

- The male offspring of affected males and all unaffected females have a 50% chance of getting the disease.
- In affected individuals, level of a protein necessary for brain cell development is approximately halved.
- The mutant allele codes for a non-functional protein as its DNA sequence contains tandem repeats within the coding region of the gene.
- Typically, symptoms do not start until after the age of 30 and the disease progressively worsens with age.
- People affected by the disease die during the age of 40 to 50.

What do these discoveries suggest about Huntington's disease?

A The allele resulting in Huntington's disease is located on the X chromosome.

B Normal brain cell development requires expression of two normal alleles to produce normal levels of the critical protein and thus the disease-causing allele is dominant.

C It is controlled by multiple genes and symptoms can be influenced by environmental factors.

D The expression of the disease-causing allele is masked by the presence of a normal allele.

18 A cross between a round-leafed, tall plant and round-leafed, dwarf plant produced the following offspring:

121 round-leafed, tall plant
124 round-leafed, dwarf plant
42 oval-leafed, tall plant
37 oval-leafed, dwarf plant

Key
R – round leaf
r – oval leaf
T – tall
t – dwarf

What were the genotypes of the parents?

A RrTt x Rrtt

B RrTt x RRtt

C RrTT x Rrtt

D RrTT x RRtt

19 In rabbits, there are two alleles for fur colour, grey and white, and two alleles for fur length, short and long. Two pure-breeding rabbits were mated, and the F_1 offspring all had grey and long hair. When the F_1 offspring were selfed, they produced the following numbers of F_2 offspring:

grey and long haired	92
grey and short haired	32
white and long haired	28
white and short haired	13

Which of the following statement(s) is/are true?

- I The genes for fur colour and fur length assort independently.
- II The probability of producing pure-bred offspring is 1 in 16.
- III The original pure-breeding parents must be only grey and long haired & white and short haired.
- IV Two out of the four phenotypes in the F_2 offspring are formed as a result of crossing over.

A I only

B II and III

C I and III

D II, III and IV

20 Fruit shape in summer squash plants is controlled by two independently segregating gene loci.

The alleles for each gene locus and the traits they code for are shown below.

A	spherical	a	long

B	spherical	b	long

When a plant has at least one copy of the dominant allele of both genes, the fruits of the plant will be disk-shaped instead.

Plants which have disk-shaped fruits are selfed and the numbers of offspring plants having different fruit shapes were recorded.

disk-shaped	354
spherical	248
long	38

Which of the following statements are true?

1 The mode of inheritance for fruit shape in summer squash is recessive epistasis.

2 Summer squash plants which are pure-breeding for fruit shape will always have either long fruits or disk-shaped fruits.

3 When two heterozygotes are crossed, the offspring phenotypic ratio will be 9 disk-shaped fruit : 6 spherical fruit : 1 long fruit.

4 The proteins coded for by the gene loci, A/a and B/b, are both involved in the same biochemical pathway that controls fruit shape development in summer squash.

A 1 and 2

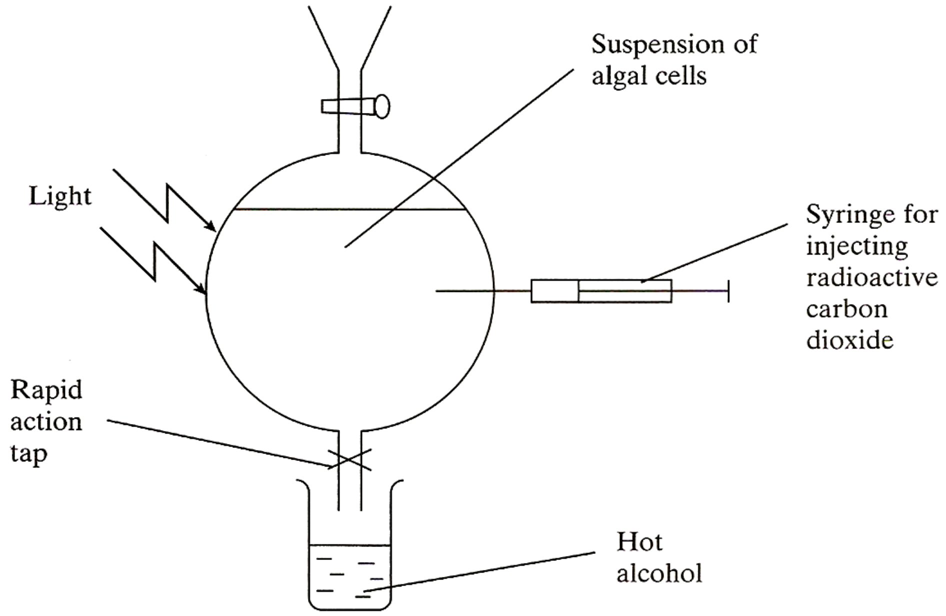
B 1 and 4

C 2 and 3

D 3 and 4

21 An investigation was carried out to find out the sequence of biochemical changes that occur during photosynthesis. Radioactive carbon dioxide was added to a suspension of algal cells, and they were exposed to light. At intervals, samples of the suspension were removed into hot alcohol.

The experimental set up is shown below.



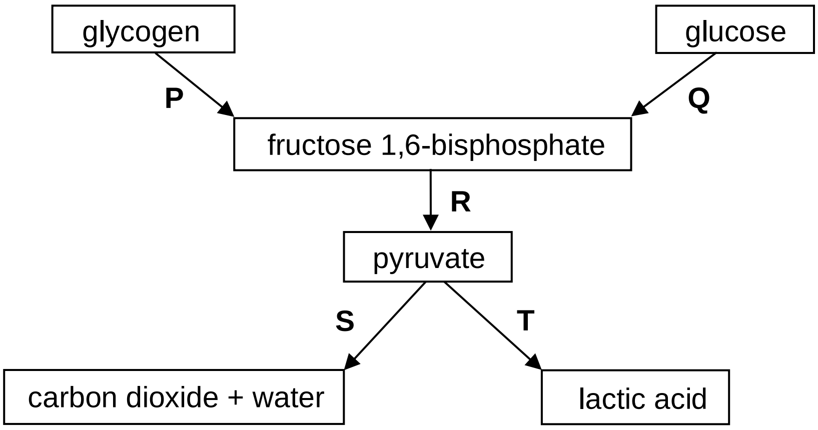
Samples were removed from the suspension at five different times, between 5 seconds and 600 seconds after the start of the experiment. These extracted samples were analysed for different radioactively labelled compounds. In each sample, the amount of radioactivity present in four different organic compounds, P, Q, R and S was measured. The table shows the results.

organic compound	amount of radioactivity present / arbitrary units				
	5 s	15 s	60 s	180 s	600 s
P	0.01	0.02	0.08	0.17	0.67
Q	1.00	2.00	3.10	3.15	3.15
R	0.10	1.50	2.20	2.30	2.40
S	0.05	0.11	0.16	1.00	1.00

What are the most likely identities of the four organic compounds?

	P	Q	R	S
A	G3P	GP	6 carbon intermediate	RuBP
B	GP	RuBP	G3P	6 carbon intermediate
C	RuBP	6 carbon intermediate	GP	G3P
D	RuBP	GP	G3P	6 carbon intermediate

22 With reference to the diagram, relate processes P, Q, R, S, T to statements I, II and III.



I NAD$^+$ is regenerated without the use of the electron transport system.
II ATP is synthesised via substrate-level phosphorylation.
III It can take place under anaerobic conditions.

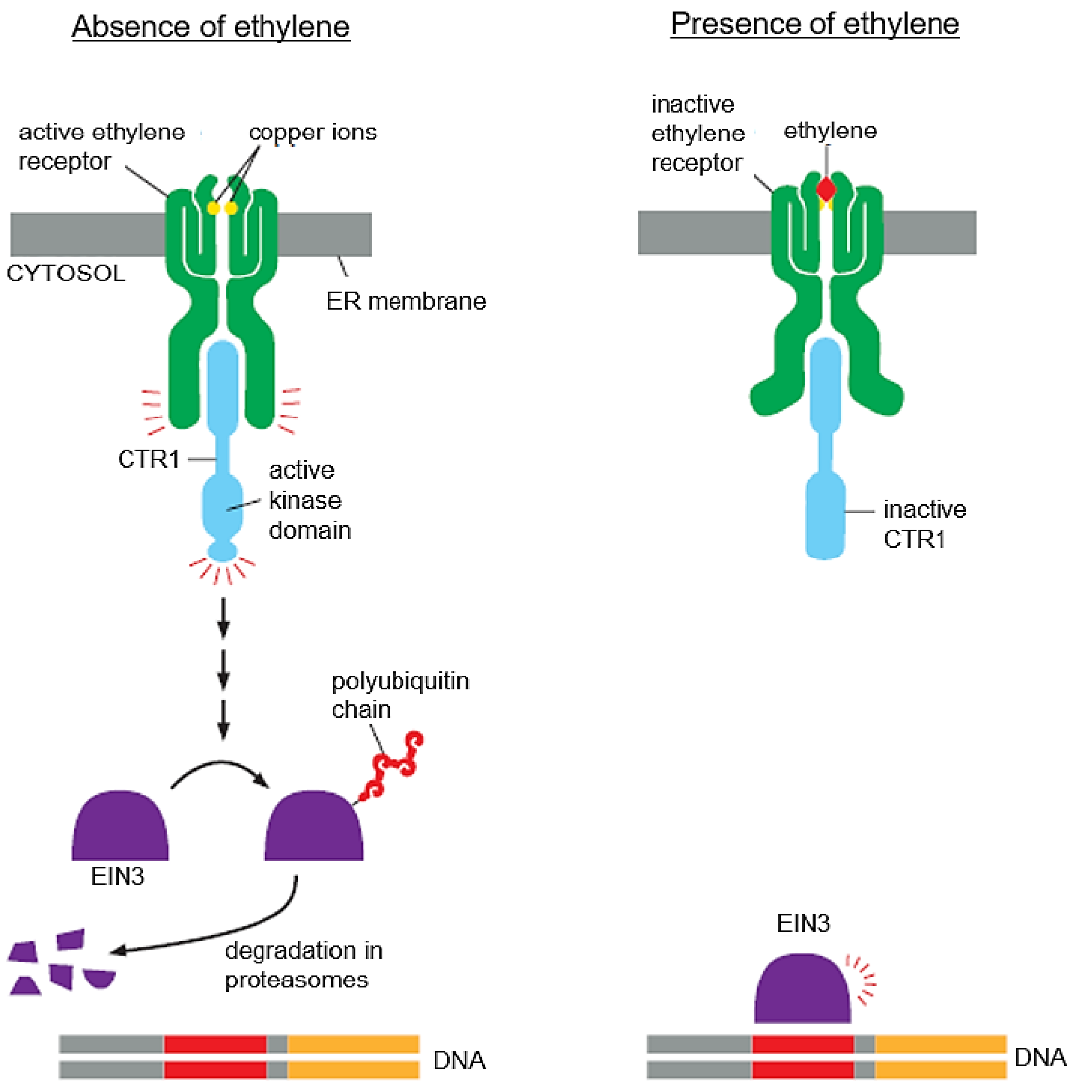
	I	II	III
A	R	S	T
B	T	R	P
C	none of the above	S	R
D	S	Q	none of the above

23 Which of the following is correct with regards to kinases and second messengers?

	kinase		second messenger	
	type of molecule	directly involved in	type of molecule	directly involved in
A	nucleotide	signal amplification	nucleotide	signal transduction
B	nucleotide	signal transduction	protein	signal amplification
C	protein	signal transduction	protein	signal amplification
D	protein	signal amplification	nucleotide	signal transduction

24 Ethylene gas is a plant hormone produced to cause the ripening of fruits.

In plants, the ethylene receptors are embedded on the endoplasmic reticulum. The ethylene signalling pathway is shown in the diagram below.



Based on the above information, which of the following statements are **incorrect**?

1 When CTR1's kinase domain is activated, signal amplification is carried out by CTR1 and fruits will ripen.
2 Ligands that bind to the ethylene receptor are hydrophilic and cannot pass through the cell surface membrane.
3 Binding of ethylene to ethylene receptor results in conformational change in the intracellular domain of the receptor and the receptor is activated.
4 Activation of EIN3 results in ripening of fruits.

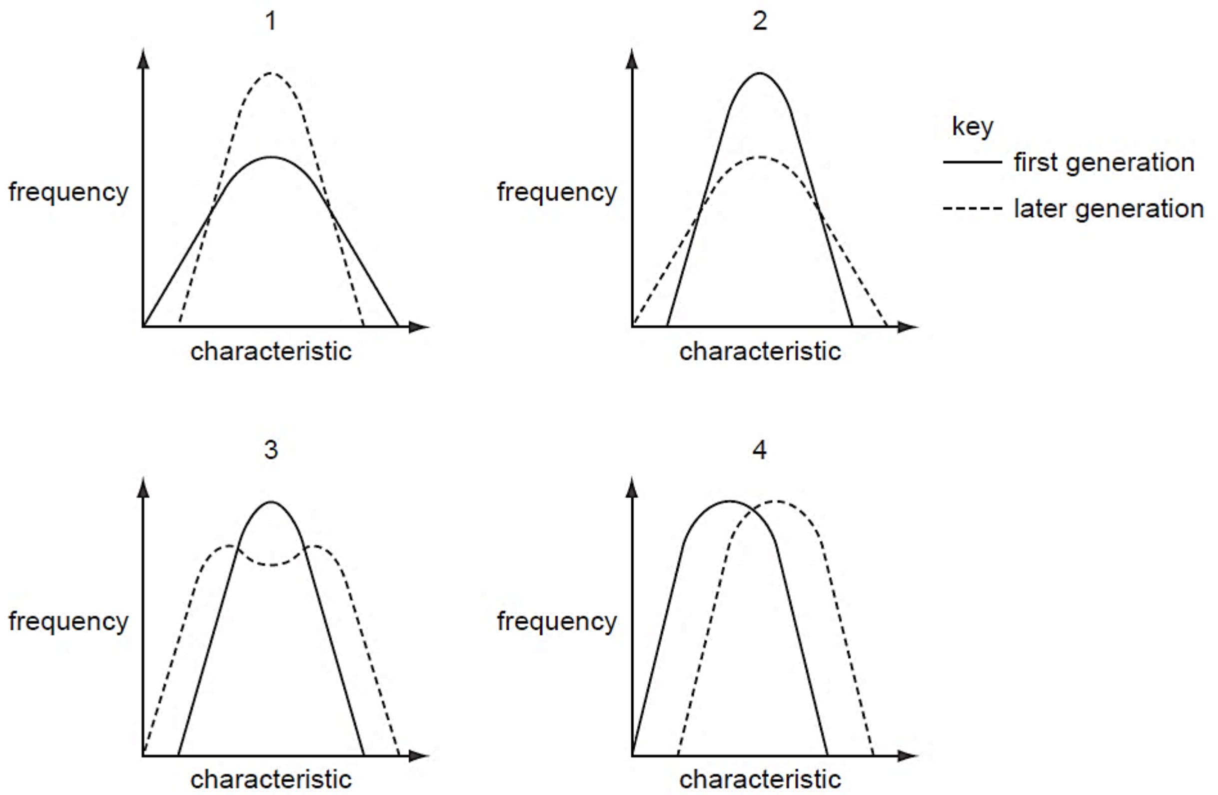
A 1 and 2

B 2 and 3

C 3 and 4

D 1, 2 and 3

25 The graphs show frequency against a measured characteristic in the first and later generation of an organism.



Which graph represents each type of natural selection?

	directional	disruptive	stabilising
A	1	2	3
B	2	3	4
C	3	1	2
D	4	3	1

26 A student recorded the shell colour and banding pattern of all empty shells of the brown-lipped snail, *Cepaea nemoralis*, found in the woods. Wide variations in shell colour and banding pattern, which are genetically controlled, was shown.

Gene **C** codes for the colour of the shell. There are six alleles.

$\mathbf{C}^{B}$	$\mathbf{C}^{DP}$	$\mathbf{C}^{PP}$	$\mathbf{C}^{FP}$	$\mathbf{C}^{DY}$	$\mathbf{C}^{PY}$
brown	dark pink	pale pink	faint pink	dark yellow	pale yellow

most dominant $\longrightarrow$ least dominant

Gene **B** codes for the presence and absence of bands. There are two alleles.

$\mathbf{B^{O}}$ = unbanded, dominant allele
$\mathbf{B^{B}}$ = banded, recessive allele

Which conclusion drawn from this information is valid?

A In the woods, brown, unbanded phenotypes will occur in the highest frequency and banded, pale yellow phenotypes will occur in the lowest frequency.

B More than one selection pressure is likely to be acting to maintain the polymorphism shown in shell colour and banding pattern of the snails.

C Predation by birds, such as song thrush, is the only selection pressure acting on *C. nemoralis* in the woods.

D The proportion of each phenotype in the population of *C. nemoralis* in the woods will be similar to the proportion of each phenotype recorded for the empty shells.

27 Which observation of a feature indicates divergence due to evolution?

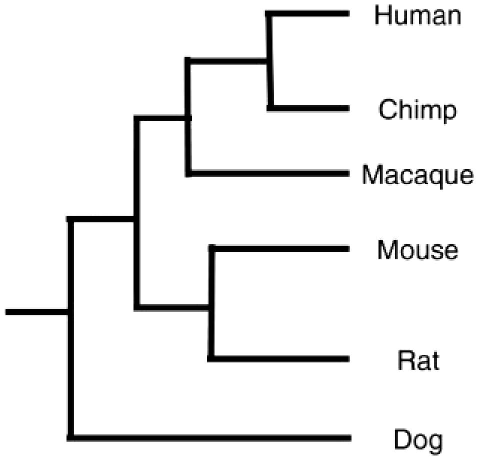
A Birds and butterflies both have wings but are distantly related. They can live in different environmental conditions.

B Birds and butterflies both have wings but are distantly related. They can live in the same environmental conditions.

C Some closely-related species of bird have different sizes and shapes of beaks. The species live in different environmental conditions.

D Some closely-related species of bird have the same size and shape of beaks. The species live in different environmental conditions.

28 The differences between the trypsin molecules from different species have been used to produce a phylogenetic tree.



Based on the above information, how many of the following statement(s) is/are true?

- Chimp and macaque share a more recent common ancestor than mouse and rat.
- A total of five speciation events are shown.
- Differences in the different levels of protein structures of trypsin between the six species was used to build the phylogenetic tree.
- Trypsin is a homologous protein.

A 1 **B** 2 **C** 3 **D** 4

29 Muscle cells have cell surface receptors for the neurotransmitter, ACh. These receptors allow the muscle to respond to a nerve impulse.

In the condition myasthenia gravis, the helper T cells stimulate a type of naïve B cells to become plasma cells and secrete antibodies that bind to and block these ACh receptors. These muscles cannot be stimulated and begin to break down.

A short-term treatment for myasthenia gravis is to inject antibodies produced by a particular plasma cell into the blood.

What is the target of injected antibodies?

A antibodies that bind to ACh receptors

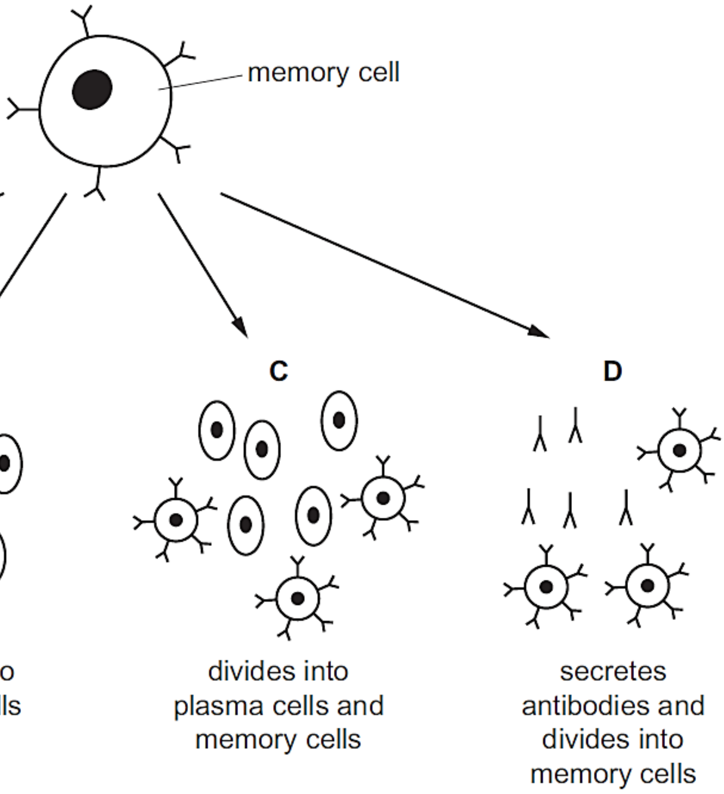
B muscle cell surface ACh receptors

C plasma cells secreting antibodies to target the nerve cells

D the stimulating neurotransmitter ACh

30 When exposed to an antigen for a second time, memory cells stimulate a secondary immune response.

Which correctly shows the secondary immune response?



Ans = **C**

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